CHEZTER P. VARGAS

Biomedical Engineering Graduate | Embedded Systems | Clinical Engineering

hire_me@cheztervargas.com | +64 27 479 9150 | Wellington 6037, New Zealand
cheztervargas.com | linkedin.com/in/cheztervargas | github.com/LeCaseus

PROFESSIONAL SUMMARY

Biomedical engineering graduate with a strong interest in embedded systems and their applications in healthcare. Experienced in sEMG-based biosignal processing and embedded systems development, with complementary skills in C/C++ programming. Seeking to contribute to R&D projects and healthcare or medical environments within a forward-thinking organization. Open to employer-sponsored visa arrangements while a residence visa application is in progress.

EDUCATION

Bachelor of Science in Biomedical Engineering **Aug 2021 – Dec 2025**
Ateneo de Zamboanga University | Zamboanga City, Philippines

- GPA: 3.28 / 4.00
- Thesis: eM-Brace, Development of an Autonomous Wrist Splint with an Integrated Massage System for Alleviating Muscle Weakness Related to Carpal Tunnel Syndrome
- Relevant coursework: Anatomy and Physiology for Engineers, Biomaterials and Biocompatibility, Biomedical Instrumentation, Biomedical Signal Processing, Health Technology Management
- Officer, Biomedical Engineering Society, Head of Creatives (2023 – 2024)

WORK EXPERIENCE

Biomedical Engineering Intern **Jun 2024 – Jul 2024**
Zamboanga City Medical Center | Zamboanga City, Philippines

- Calibrated and functionally tested clinical equipment using Fluke patient simulators, defibrillator analyzers and medical gas flow analyzers.
- Disassembled, cleaned and recalibrated portable devices, including medical gas flow meters and fingertip pulse oximeters, for redeployment to hospital departments.
- Performed in-situ preventive maintenance on fixed equipment such as patient monitors directly at ward locations using portable calibration tools.
- Completed and filed work orders to ZCMC BME departmental standards, maintaining accurate maintenance histories.
- Performed equipment rounds across hospital departments to identify and schedule equipment due for maintenance from the master list.
- Completed a rotation in the ENT Medical Makerspace covering FDM 3D printing, slicer workflows and basic CAD; designed a phone stand and printed three models on department printers.

PROJECTS

eM-Brace: Autonomous Carpal Tunnel Syndrome Rehabilitation Device **2024 – 2025**
Undergraduate Thesis Project | Ateneo de Zamboanga University

- Designed and developed an autonomous wrist splint with an integrated myofascial release massage system to alleviate muscle weakness associated with Carpal Tunnel Syndrome (CTS).
- Detected wrist rest state from EMG electrodes on the abductor pollicis brevis using real-time signal acquisition, processing and RMS threshold classification, achieving 100% classification accuracy in functional testing.
- Massage module delivers 1.36 kPa of therapeutic pressure and activates autonomously only during detected muscle rest.
- Auto-Fit module adjusts splint tension to maintain wrist support pressure of about 3.68 kPa across wrist sizes; Motion Support module loosens the strap during active states, allowing up to 10° flexion and 30° extension.
- Beta tested on 15 participants with induced CTS symptoms: 73% improved grip strength by more than 2.2 kg, and mean pain scores fell from 3.27 to 0.33.
- Platform: ESP32 WROOM. Signal pipeline: analogue signal conditioning, notch filtering, digital reconstruction and threshold-based classification. Evaluated by a licensed physical therapist.

Personal Website and Self-Hosted Server **2026**
cheztervargas.com | Personal project

- Built and self-host a personal website on a home Linux laptop: static site generated with Hugo, served through Caddy and a Cloudflare Tunnel.
- Includes a small C++ backend (cpp-httplib, SQLite) handling guestbook messages, comments, reactions and webmentions, with a moderation queue.
- Developed with AI coding assistance as a learning project in C++ and web development; I deploy and maintain it myself.

SKILLS

Embedded Systems: ESP32; sEMG signal acquisition and processing; sensor integration; analogue signal conditioning

Biomedical: Calibration and preventive maintenance; Fluke patient simulators; defibrillator analyzers; medical gas flow analyzers; fingertip pulse oximeters

Programming: C/C++ (basic); HTML/CSS/JavaScript (beginner)

Self-Hosting: Linux; Hugo; Caddy; Podman; Cloudflare Tunnel; SQLite

Tools and Software: Git; Linux; VS Code; Arduino IDE; MATLAB; BSL Analysis; FDM 3D printing and CAD (basic)

Clinical Documentation: Work orders; equipment maintenance records; master equipment lists; departmental reporting

Professional: Analytical problem-solving; teamwork; adaptability; time management

CERTIFICATIONS AND TRAINING

- On-The-Job Training Certificate, Zamboanga City Medical Center (2024)
- PCB Development and Design Workshop, Junior IECEP ADZU Chapter (2023)
- Youth Volunteer Orientation Course, Philippine Red Cross (2019)

HONOURS AND AWARDS

- Most Diligent Intern, Zamboanga City Medical Center (2024)
- Best in Calibration, Zamboanga City Medical Center (2024)
- Best Product Demo Performance, ENT-HNS Medical Makerspace (2024)
- Jack of All Trades Award, ENT-HNS Medical Makerspace (2024)

VOLUNTEER AND EXTRACURRICULAR

Head of Creatives **2023 – 2024**
Biomedical Engineering Society | Zamboanga City, Philippines

Creatives Team Member **2021 – 2022**
SITEAO Academic Organization | Zamboanga City, Philippines

Public Information Officer **2019 – 2020**
Senior Plus Red Cross Youth | Zamboanga City, Philippines

LANGUAGES

English: Fluent | Tagalog: Native | Chavacano (Spanish-based creole): Intermediate

REFERENCES

Available upon request.